

OCR, OMR, MICR — Detailed Lecture Notes

JKSSB Computer Awareness

October 8, 2026

13.1 Reading machines and why they matter

A computer works only with data that has already been converted into machine-readable form. When the data exists on paper — as printed text, as a filled-in bubble on a form, or as a line of characters on a cheque — the machine needs a way to read it. A **reading machine** is the device or technology that captures this physical information and turns it into digital data the system can process. The physical reader (often a scanner, sometimes a dedicated reader) performs the capture; the *recognition method* decides what the captured pattern means. This lesson covers the three most examined recognition methods — OCR, OMR, and MICR — together with the code-and-tag devices (barcode, QR code, RFID) and biometric input, because questions frequently ask candidates to tell these near-neighbours apart.

The single most important idea is the *read-target* of each method. **OCR** reads *text*. **OMR** reads *marks*. **MICR** reads *characters printed in magnetic ink*. Almost every item on this topic can be solved by fixing this one distinction in memory.

All the devices in this lesson are *input* devices or reading *technologies*: they bring data into the system. Most of them are also *direct* input, because the device reads the object, mark, code, or trait itself, instead of asking the user to type it. This is why a reading machine is faster and less error-prone than a keyboard for a repetitive task such as reading a thousand answer sheets or a thousand cheques.

Definition (Reading machine). *A device or technology that captures printed characters, marks, codes, or tags from a physical medium and converts them into machine-readable data for the computer to process.*

13.2 Terminology at a glance

The table below fixes the meaning of every term used in this lesson. Read it down once; then, for each term, be ready to state the read-target and one place it is used.

Term	Meaning in this lesson
OCR	Optical Character Recognition — converts an image of printed or written text into editable text.
OMR	Optical Mark Recognition — detects the presence or absence of marks at fixed positions on a form.
MICR	Magnetic Ink Character Recognition — reads characters printed in magnetic ink, used on bank cheques.
Barcode	A one-dimensional pattern of parallel lines read optically to identify an item.
QR code	A two-dimensional matrix of squares (Quick Response) that holds more data than a barcode.
RFID	Radio Frequency Identification — a tag and reader exchange radio waves; no line of sight is needed.
Biometrics	Input that identifies a person from a physical or behavioural trait (fingerprint, iris, face).

13.3 OCR — Optical Character Recognition

OCR stands for **Optical Character Recognition**. It takes a scanned image of *printed or handwritten text* and converts it into machine-encoded, *editable text*. The key word is *character*: OCR recognises individual letters, digits, and symbols, so the output is text that a computer can search, copy, and edit. This is what makes OCR different from OMR, which recognises only whether a position has been marked and never identifies a letter.

OCR is used to digitise the printed word. A book, a bundle of invoices, a passport page, or an old form can be scanned into an image, and OCR software then extracts the text from that image. A useful way to remember the pairing is: the **scanner** is the hardware that creates the image, while **OCR** is the technology that turns the image of text into actual, selectable text.

Definition (OCR). *A recognition technology that reads characters from a scanned image of printed or handwritten text and converts them into machine-encoded, editable text.*

OCR works best on clean, clearly printed text in a known font; its accuracy drops when the page is skewed, smudged, or handwritten. When the input is handwritten rather than printed, the same idea is sometimes called *Intelligent Character Recognition* (ICR), but for exam purposes both count as OCR-style reading of text. OCR is *not* a storage device and *not* a printer: it only converts an existing image into text.

Example 13.3.1. *A student scans a page of a textbook into the computer. The scanner produces an image, so the words on the page cannot yet be copied or searched. Running OCR on that image produces a text file in which every word can be selected and edited. The scanner supplied the image; OCR supplied the text.*

13.4 OMR — Optical Mark Recognition

OMR stands for **Optical Mark Recognition**. It reads a form by detecting the *presence or absence of a mark* at fixed positions. Typically the candidate fills a small bubble or shades a box with a pencil or pen, and the OMR reader checks which positions have been darkened. Crucially, OMR reads *marks*, not characters: it does not know what a letter “says”, only that a particular position has been filled.

Because it detects marks at fixed coordinates, OMR is ideal for forms where the answer is one of a small set of choices. Its classic use is the **MCQ answer sheet**: the machine reads which bubble has been shaded for each question. It is also used for questionnaires, surveys, census forms, and ballots. A very common exam trap is to confuse OMR with OCR; remember that OMR works on *marks*, while OCR works on *text*.

Definition (OMR). *A recognition technology that detects the presence or absence of marks at fixed positions on a form, used mainly to read objective (MCQ) answer sheets and survey forms.*

OMR is extremely fast because it does not have to interpret a shape as a character; it only has to decide whether a small region is dark or light. That speed is why OMR is the standard method for large-scale examinations where millions of answer sheets must be processed quickly. Its natural limit is that it can record only *which* option was marked, not any written answer: if the candidate writes a sentence instead of shading a bubble, OMR cannot read it.

13.5 MICR — Magnetic Ink Character Recognition

MICR stands for **Magnetic Ink Character Recognition**. It reads characters printed in a special *magnetic ink* that contains iron oxide. The characters are printed in a standard font along the bottom of a *bank cheque* as the **MICR line**, which normally encodes the cheque number, the bank and branch code, and the account number. A MICR reader magnetises the ink and detects the shapes of the characters as they pass the read head.

Banking is the defining application of MICR. Cheque processing must be very fast and highly secure, and MICR satisfies both needs: the reading is quick, and because the ink is magnetic it is difficult to forge or to alter without detection. When a question mentions a *bank cheque* or *magnetic ink*, the answer is almost always MICR.

Definition (MICR). *A recognition technology that reads characters printed in magnetic ink, used mainly in banking to process the MICR line printed at the bottom of a cheque.*

Unlike ordinary ink, magnetic ink can be read even if the cheque is slightly crumpled or over-stamped, because the reader senses the magnetic field of the ink rather than a purely visual image. This ruggedness and the difficulty of counterfeiting magnetic ink are exactly the reasons banks adopted MICR for cheque clearing.

13.6 The crucial trio: OCR vs OMR vs MICR

These three names are the heart of the lesson, and JKSSB questions routinely swap their meanings in the options. The table below is the single most useful thing to memorise.

Method	What it reads	Where it is used
OCR	Printed or written characters	Digitising books, documents, and passports
OMR	Marks on a form	MCQ answer sheets, surveys, ballots
MICR	Characters in magnetic ink	Bank cheques and cheque clearing

The distinction to hold is: **MICR** uses *magnetic ink*; **OCR** reads *printed text*; **OMR** reads *marks*. This is the JKSSB trap TRAP-25, and the real paper item behind it (PYQ-30) asks simply that magnetic ink means MICR.

It also helps to picture *how* each one captures its data. OMR and OCR both read an optical (light) signal: the reader sees the paper. MICR is the odd one out, because its reader senses a *magnetic* signal from the ink. So a quick way to separate them in a hurry is: two are *optical*, one is *magnetic*.

Method	Signal read	Input pattern	Cannot do
OCR	Optical (light)	Text characters	Cannot read marks alone
OMR	Optical (light)	Marks / shaded bubbles	Cannot read written characters
MICR	Magnetic	Magnetic-ink characters	Not used for forms or plain text

13.7 Barcodes and QR codes

A **barcode** is a one-dimensional (1D) pattern of parallel lines of varying width and spacing. A barcode scanner shines a light beam across the lines and reads the reflected pattern, which represents a number or code that identifies a product. Because the scanner must see the lines, a barcode requires *line of sight*. Barcodes are used on product labels, on the ISBN of books, and on postal items.

A **QR code** — **Quick Response** code — is a *two-dimensional* (2D) matrix of black and white squares. It stores more data than a barcode and can be read from any direction. Because it can hold a URL, scanning a QR code can open a website or an app directly. Both a barcode and a QR code are read *optically*, so the nearest-neighbour trap is simply that a QR code is two-dimensional and carries more information, while a barcode is one-dimensional.

Definition (QR code). *A two-dimensional matrix of black and white squares (Quick Response code) that stores more data than a barcode and can be read from any direction.*

Example 13.7.1. *A shop assistant passes a product over a laser at the billing counter: the scanner reads the 1D barcode on the packet and looks up its price. A customer points a phone camera at the square QR code on a poster: the phone reads the 2D code and opens the shop's website. The barcode carried a short number; the QR code carried a whole web address.*

13.8 RFID — Radio Frequency Identification

RFID stands for **Radio Frequency Identification**. An RFID system has two parts: a *tag* attached to the object and a *reader* that interrogates it. The two exchange *radio waves*, so no line of sight is required, unlike a barcode. A passive tag carries no battery and is powered by the reader's signal; an active tag has its own battery and can broadcast over a longer range.

RFID is used where the tag may be out of sight or where many tags must be read quickly: toll collection (for example FASTag), access-control cards, passports, library books, and inventory tracking. The exam contrast is always barcode/QR versus RFID: barcode and QR are read *optically* and need line of sight, while RFID uses *radio waves* and can read a tag without it being visible.

13.9 Biometric input

Biometrics identifies a person from a physical or behavioural trait. The common physical traits are the **fingerprint**, the **iris**, the **face**, and the **palm**; the voice is a common behavioural trait. A

sensor captures the trait and feeds the measured data to the computer as *input*; the system then compares it with the stored template. Biometric input is a *direct* form of input because the sensor reads the trait itself rather than asking the user to type or encode it.

Biometrics are used for attendance systems, device unlocking, and authentication, and they appear in national identity and benefit systems. A useful checklist item for the exam is that a fingerprint reader, an iris scanner, and a face-recognition camera are all *input* devices; a device that merely prints or displays the result is *output*. Do not confuse biometric traits (fingerprint, iris, face) with the optical reading machines: a fingerprint sensor reads a part of the human body, not a paper code.

13.10 Full-form ladder and memory hooks

Direct-recall items on this topic usually ask for a full form, so learn the expansions exactly. The same expansions appear on the slides; repeat them until they can be produced without hesitation.

Short form	Full form
OCR	Optical Character Recognition
OMR	Optical Mark Recognition
MICR	Magnetic Ink Character Recognition
QR	Quick Response
RFID	Radio Frequency Identification

Three one-line hooks carry most of the content of this lesson. First, *OCR reads text, OMR reads marks, MICR reads magnetic ink*. Second, *a barcode is one-dimensional, a QR code is two-dimensional, and RFID uses radio waves*. Third, *a biometric sensor reads a human trait — fingerprint, iris, face, or voice*. If a question seems unfamiliar, reduce it to one of these three sentences and the answer usually becomes clear.

13.11 Applications at a glance

The table gathers the reading machines and code devices with their defining application, for quick revision.

Technology	Defining application
OCR	Converting scanned printed text into editable text
OMR	Reading marks on objective (MCQ) answer sheets
MICR	Reading the magnetic-ink line at the bottom of a bank cheque
Barcode	Reading 1D product labels and book ISBNs
QR code	Reading 2D codes that can open a link or store more data
RFID	Reading tags by radio waves (tolls, access cards, inventory)
Biometrics	Identifying a person from a fingerprint, iris, face, or voice

13.12 Exam retrieval and common confusions

Six distinctions prevent most errors on this topic.

1. **MICR** uses **magnetic ink**; **OCR** reads **printed text**; **OMR** reads **marks**. Do not interchange them (TRAP-25, PYQ-30).
2. A real PYQ asks which technology uses magnetic ink on cheques: the answer is **MICR** (PYQ-30).
3. **OMR**, not OCR, is used for **MCQ answer sheets**, because it reads marks at fixed positions.
4. A **QR code** is two-dimensional and holds more data than a one-dimensional **barcode**; both are read optically.
5. **RFID** uses **radio waves** and needs no line of sight; a barcode or QR code must be *seen* by the reader.
6. A **fingerprint** or **iris** scanner is **input**; biometrics identify a person from a physical or behavioural trait.

Example 13.12.1. *An item states that a bank reads the characters at the bottom of a cheque using OCR. This is wrong. The cheque characters are printed in magnetic ink, so the technology is **MICR**. OCR would be used to read ordinary printed text from a scanned document, and OMR would be used to read marks on a form — neither of which matches a cheque’s magnetic-ink line.*

Example 13.12.2. *An item lists four reading methods and asks which two are optical. OCR and OMR both use a light signal to read the paper, so they are the two optical methods. MICR is magnetic, not optical, and RFID is neither optical nor magnetic — it uses radio waves. Separating the methods by their signal (optical, magnetic, radio) is a fast way to answer such a question.*

Exercises

1. Expand OCR, OMR, and MICR, and state in one line what each reads.
2. Which technology would read an objective (MCQ) answer sheet, and why?
3. Explain why a bank uses MICR rather than OCR for cheque processing.
4. Distinguish a barcode from a QR code on two points.
5. State one way in which RFID differs from a barcode.
6. Give three examples of biometric traits and state the purpose for which biometric input is commonly used.
7. Classify these as optical, magnetic, or radio reading: OMR, MICR, RFID, OCR.

Solutions

1. OCR = Optical Character Recognition, which reads printed or written characters; OMR = Optical Mark Recognition, which reads marks on a form; MICR = Magnetic Ink Character Recognition, which reads characters in magnetic ink.
2. OMR, because it detects the presence or absence of a mark at fixed positions, which is exactly how a candidate fills bubbles on an MCQ answer sheet.
3. The cheque characters are printed in magnetic ink, which MICR reads directly; the reading is fast, and the magnetic ink is difficult to forge or alter. OCR reads ordinary printed text, not magnetic-ink characters.

4. A barcode is a one-dimensional pattern of parallel lines and holds less data; a QR code is a two-dimensional matrix of squares, holds more data, and can be read from any direction.
5. RFID exchanges radio waves between a tag and a reader and needs no line of sight, whereas a barcode is read optically and must be visible to the scanner. RFID can also read many tags at once.
6. Fingerprint, iris, and face (voice and palm are also valid). Biometric input is commonly used for attendance, authentication, and device unlocking.
7. OMR and OCR are *optical*; MICR is *magnetic*; RFID is *radio*.